EXPERIMENT-2

Design a database for a College Admission System. Create Student and Department tables and apply appropriate integrity constraints to ensure valid and consistent data. Demonstrate all six constraints with suitable SQL statements

S.No	Requirement	Constraint
1	Uniquely identify each student	PRIMARY KEY
2	Establish relationship between Student and Department	FOREIGN KEY
3	Ensure email ID is unique	UNIQUE
4	Validate student age/marks	CHECK
5	Assign a default student status	DEFAULT
6	Ensure student name is mandatory	NOT NULL

Aim:

To design a database for a College Admission System by creating Student and Department tables and applying six integrity constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, and NOT NULL.

Step 1: Create the Department Table

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50) NOT NULL
);
```

Output – Department table created

Dept_ID Dept_Name

(empty)

Step 2: Create the Student Table with Six Constraints

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    FirstName VARCHAR(30) NOT NULL,
    LastName VARCHAR(30) NOT NULL,
```

```
    Email VARCHAR(100) UNIQUE,
    Age INT CHECK (Age >= 17 AND Age <= 30),
    Marks INT CHECK (Marks >= 0 AND Marks <= 100),
    Status VARCHAR(20) DEFAULT 'Pending',
    Dept_ID INT,
    FOREIGN KEY (Dept_ID)
        REFERENCES Department(Dept_ID)
);
```

Constraints applied:

- PRIMARY KEY – uniquely identifies each student.
- FOREIGN KEY – connects Student with Department.
- UNIQUE – prevents duplicate email addresses.
- CHECK – validates age and marks.
- DEFAULT – assigns Pending when status is omitted.
- NOT NULL – makes first and last names mandatory.

Step 3: Insert Values into Department

```
INSERT INTO Department VALUES
(1, 'CSE'),
(2, 'ECE'),
(3, 'IT');
```

Output – Department table

Dept_ID	**Dept_Name**
1	CSE
2	ECE
3	IT

Step 4: Insert Values into Student

```
INSERT INTO Student
(Student_ID, FirstName, LastName, Email, Age, Marks, Dept_ID)
VALUES
(101, 'Arun', 'Kumar', 'arun@gmail.com', 20, 85, 1),
(102, 'Priya', 'Devi', 'priya@gmail.com', 19, 90, 2),
(103, 'Kumar', 'Raj', 'kumar@gmail.com', 21, 78, 3);
```

Since Status is omitted, the default value Pending is assigned automatically.

Step 5: Display the Student Table

```
SELECT * FROM Student;
```

Output – Student table

Student_ID	FirstName	LastName	Email	Age	Marks	Status	Dept_ID
101	Arun	Kumar	arun@gmail.com	20	85	Pending	1
102	Priya	Devi	priya@gmail.com	19	90	Pending	2
103	Kumar	Raj	kumar@gmail.com	21	78	Pending	3

Step 6: Demonstrate the Six Constraints

Run each test separately after completing the previous steps. The invalid examples are expected to produce constraint errors.

1. PRIMARY KEY

```
INSERT INTO Student
VALUES (101, 'Ravi', 'Sharma', 'ravi@gmail.com',
    20, 80, 'Pending', 1);
```

2. FOREIGN KEY

```
INSERT INTO Student
```

```
VALUES (104, 'Ravi', 'Sharma', 'ravi@gmail.com',
    20, 80, 'Pending', 99);
```

3. UNIQUE

```
INSERT INTO Student
VALUES (104, 'Ravi', 'Sharma', 'arun@gmail.com',
    20, 80, 'Pending', 1);
```

4. CHECK

```
INSERT INTO Student
VALUES (104, 'Ravi', 'Sharma', 'ravi@gmail.com',
    15, 80, 'Pending', 1);
```

5. DEFAULT

```
INSERT INTO Student
(Student_ID, FirstName, LastName, Email, Age, Marks, Dept_ID)
VALUES (104, 'Ravi', 'Sharma', 'ravi@gmail.com', 22, 88, 1);

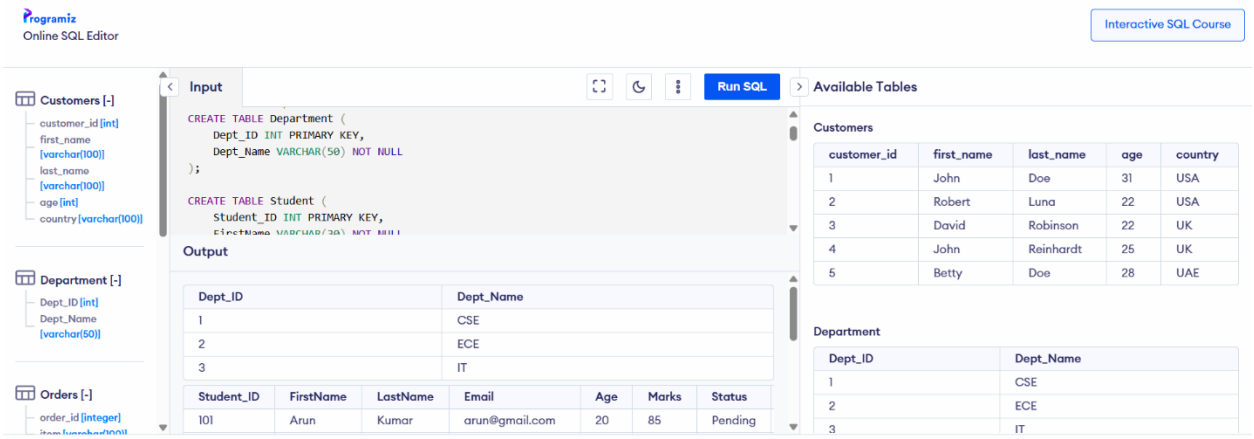
SELECT * FROM Student;
```

6. NOT NULL

```
INSERT INTO Student
VALUES (105, NULL, 'Sharma', 'test@gmail.com',
    20, 80, 'Pending', 1);
```

Result

The Student and Department tables were created successfully. All six integrity constraints were applied to maintain data accuracy, validity, and consistency in the College Admission System.



Output

Student_ID	FirstName	LastName	Email	Age	Marks	Status
103	Kumar	Raj	kumar@gmail.com	21	78	Pending
101	Arun	Kumar	arun@gmail.com	20	85	Pending
102	Priya	Devi	priya@gmail.com	19	90	Pending
103	Kumar	Raj	kumar@gmail.com	21	78	Pending
104	Ravi	Sharma	ravi@gmail.com	22	88	Pending

Output

Student_ID	FirstName	LastName	Email	Age	Marks	Status
102	Priya	Devi	priya@gmail.com	19	90	Pending
103	Kumar	Raj	kumar@gmail.com	21	78	Pending
104	Ravi	Sharma	ravi@gmail.com	22	88	Pending
101	Arun	Kumar	arun@gmail.com	20	85	Approved
102	Priya	Devi	priya@gmail.com	19	90	Pending
103	Kumar	Raj	kumar@gmail.com	21	78	Rejected